



Addis Ababa University
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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

Medicinal Chemistry

2nd Year Assignment

Group Two

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Acknowledgement

We would like to express our sincere gratitude to our instructor Yitagesu for providing this insightful assignment on expectorants in medicinal chemistry. His guidance have been invaluable in understanding the complex mechanisms and challenges associated with developing effective treatments for respiratory conditions. This assignment has sparked a deep interest in the chemical complexities of respiratory health and the challenges of developing safe and effective medications.

Expectorants

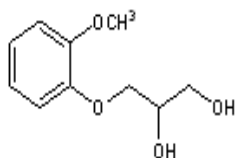
Expectorants: These are drugs that help to loosen and remove mucus from the respiratory system. They're often used to treat coughs and other respiratory conditions.

Mechanisms: Most expectorants work by one of two main mechanisms:

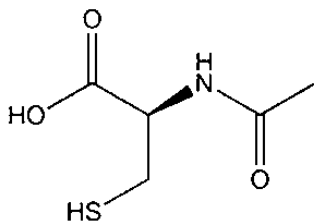
- **Increasing mucus secretion:** Think of this like thinning out the mucus to make it easier to cough up. This is often achieved by irritating the respiratory tract lining, which triggers more mucus production.
- **Decreasing mucus viscosity:** This is where the medicinal chemistry gets interesting. Some expectorants actually change the chemical structure of mucus, making it less sticky and easier to expel.

Chemical Structures: Expectorants can have a wide range of chemical structures. Some common examples include:

***Guaifenesin*:** This is a classic expectorant that works by increasing mucus secretion.



***N-acetylcysteine (NAC)*:** This is a more specialized expectorant that acts as an antioxidant and helps break down mucus.



In a nutshell: Expectorants are drugs that make it easier to cough up mucus. Medicinal chemistry helps us understand how these drugs work at the molecular level, allowing us to design and develop more effective treatments for respiratory problems.

Sedative Expectorants

Sedative expectorants are a specific class of pharmaceutical agents designed to alleviate cough and facilitate the clearance of mucus from the respiratory tract. These medications play a crucial role in managing various respiratory conditions characterized by mucus buildup and irritation. By enhancing the secretion of a protective mucous film that covers inflamed membranes, sedative expectorants improve the efficiency of mucus removal through coughing.

Sedative expectorants may be further sub-divided into three different classes :

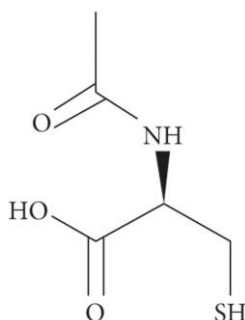
- a) ***Saline Expectorants***: These usually enhance bronchial secretion and help to “loosen” the cough, e.g., ammonium carbonate, ammonium chloride, ammonium acetate, alkali citrates (Na or K) and inorganic iodides (KI or NaI).
- b) ***Nauseant Expectorants***: These act as expectorants in small doses and nauseant and emetic in large doses, e.g., tartar emetic, ipecac, etc. These are usually mixed with sweet-tasting cough syrups that help to cure croupous bronchitis in children.
- c) ***Demulcent Expectorants***: These agents are normally mucilaginous in nature and serve to coat and protect the mucous membrane of the upper respiratory tract, e.g., syrup of acacia, ginger, glycyrrhiza (liquorice), tolubalsam and the like.

Key Examples of Sedative Expectorants include:

1. ***Acetylcysteine (N-Acetyl-L-cysteine, Mucomyst®)***

Acetylcysteine reduces the viscosity of pulmonary secretions by breaking disulfide bonds within mucus, thereby facilitating its clearance. It is commonly used as an adjuvant in bronchopulmonary disorders requiring mucolysis and acts as an antidote for acetaminophen overdose.

The structure of acetylcysteine, known under various naming conventions such as INN (International Nonproprietary Name), BAN (British Approved Name), and USAN (United States Adopted Name), remains consistent across these identifiers.



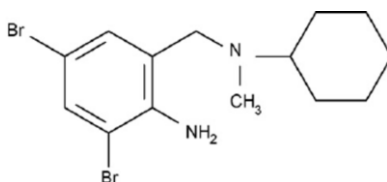
- Structure-Activity Relationship (SAR): The presence of the thiol (-SH) group is critical for its mucolytic activity. The acetylation reduces the reactivity of the sulfhydryl group while maintaining its ability to cleave disulfide bonds, thus enhancing mucus clearance.

- Dosage: Administered as 1 to 2 ml of a 10% or 20% solution every 1 to 4 hours, or by nebulization of a 20% solution (3 to 5 ml) daily.

2. *Bromhexine Hydrochloride*

Bromhexine alters the structure of bronchial secretions by increasing volume and reducing viscosity, thus enhancing mucus clearance. It also promotes the hydrolysis of mucopolysaccharides in mucus.

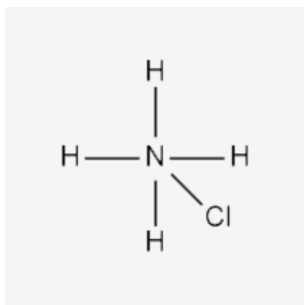
Bromhexine hydrochloride is a mucolytic agent commonly used to help with respiratory conditions. Here's the structural information:



- Structure-Activity Relationship (SAR): The bromine substituents at the 3 and 5 positions are essential for the drug's efficiency in mucus reduction. The combination of the aromatic system with the amine contributes to its biological activity, as it influences the drug's binding to mucus.
- Dosage: Adults typically take 8 to 16 mg three to four times daily; for children under 5 years, 4 mg twice daily; for children aged 5 to 10 years, 4 mg four times daily.

3. *Ammonium Chloride (NH₄Cl)*

As a saline expectorant, ammonium chloride promotes mucus secretion by facilitating a net loss of extracellular fluid and altering bicarbonate buffering in the body. This results in the secretion of less viscous mucus, aiding in symptomatic relief. Ammonium chloride (NH₄Cl) is an inorganic compound that consists of ammonium ions and chloride ions. Here's the structural information:



Ammonium chloride typically forms as a crystalline solid and is used in various applications, including as a nitrogen source in fertilizers, in food processing, and in pharmaceuticals. If you have any more questions or need additional information, just let me know!

- Structure-Activity Relationship (SAR): The quaternary ammonium structure is critical for its saline property. The ionic nature of the NH₄⁺ ion allows it to influence fluid balance in the respiratory tract.
- Dosage: Administered in doses of 300 mg to 1 g, depending on the formulation.

4. *Prepared Ipecacuanha*

Prepared ipecacuanha, a medicinal preparation, is derived from the plant *Cephaelis ipecacuanha*. It is used as an expectorant in small doses, it promotes secretions through its emetic properties at higher doses, making it effective for respiratory conditions. Active Components include alkaloids such as emetine and cephaeline .

- Structure-Activity Relationship (SAR): The presence of alkaloids with a suitable nitrogen-containing structure is necessary for its expectorant properties. The structural arrangement of these alkaloids influences their potency and safety profile.

- Dosage: Typical dosage ranges from 25 to 100 mg, delivering approximately 0.5 to 2 mg of total alkaloids.

5. *Liquorice (Glycyrrhiza)*

Liquorice is glycoside composed of glycyrrhetic acid and two units of glucuronic acid. Liquorice serves as both a demulcent and expectorant. Its soothing effects on the throat and cough suppression capabilities make it a common ingredient in cough mixtures.

- Structure-Activity Relationship (SAR): The glycyrrhizin component is crucial for its pharmacological effects. The structural characteristics of glycyrrhizin contribute to its sweetness and demulcent properties, providing a soothing effect on mucous membranes.

- Dosage: For liquorice extract, the dosage typically ranges from 0.6 to 2 g.

6. *Cocillana*

Cocillana which is from the plant *Guarea rusbyi* similar to ipecacuanha, cocillana has expectorant properties, making it useful in cough mixtures.

- Structure-Activity Relationship (SAR): The presence of specific alkaloids and glycosides in the bark determines its expectorant properties. The structural features that allow for solubility in alcohol enhance its efficacy.

- Dosage: Administered as a liquid extract in doses ranging from 0.5 to 1 ml.

7. *Potassium Iodide (KI)*

Potassium iodide acts as an expectorant, promoting mucus clearance, and has applications including treatment for cutaneous lymphatic sporotrichosis.

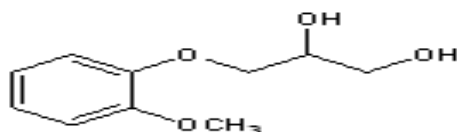
- Structure-Activity Relationship (SAR): The iodide ion's size and charge are fundamental in its role in stimulating mucus production and enhancing expectoration. Its ionic character allows for easy integration into biological systems.

- Dosage: Typically administered in the range of 250 to 500 mg.

Stimulant (Irritant) Expectorants

Drugs in this category usually induce healing in chronic inflammatory processes of the mucous membranes of the respiratory tract. These mostly comprise of mildly irritating volatile terpenoid oils and phenolic compounds based on creosote that may be inhaled and augment repair in the inflamed areas of the bronchi.

A. *Guaifensin (Mucinex)*

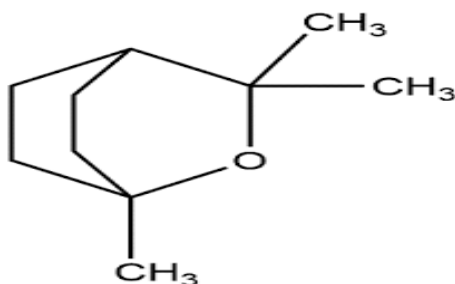


Guaifensin finds its extensive use as an expectorant. It is reported to lower the viscosity of the tenacious secretions by enhancing the volume of the respiratory tract fluid.

The action of guaifensin essentially makes better the 'dry unproductive cough' by distinctly lowering the sputum viscosity; difficulty encountered in expectoration and enhancing the sputum volume considerably. However, experimentally, it simply augments the respiratory tract secretions significantly; and that too when given in doses much higher than those usually employed therapeutically.

Guainfensin typically has safe if taken as directly. Some side effects are dizziness, constipation, headache, tiredness, rash and etc..

B. *Eucalyptol (Cineol; Cajeputol)*



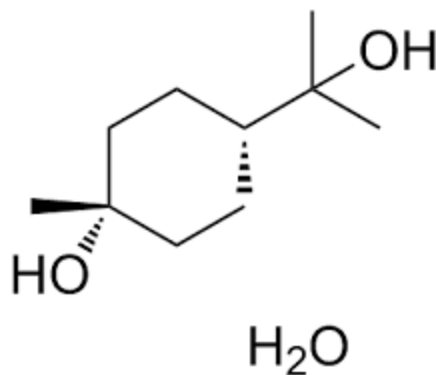
It is used locally for its antiseptic action in inflammations of nose and throat. It is also used by inhalation in bronchitis. The ‘drug’ exerts its ‘antiseptic effect’ in the inflammations of the throat and nose. It is quite often employed by inhalation in bronchitis. It is Separated from the purified volatile oil.

Allergies and respiratory failure

C. *Terpin Hydrate*

A humectant and expectorant, terpin hydrate works directly on the bronchiale secretory cells in the lower respiratory tract to liquify and facilitate the elimination of bronchial secretions. It also exerts a weak antiseptic effect on the pulmonary parenchyma. It is derived from sources such as turpentine, oregano, thyme, and eucalyptus.

It has been duly observed that the ‘terpin hydrate elixir’ as such normally contains too little of the active compound to make it effective single ; and, therefore it is invariably used mainly as a vehicle for the cough mixtures, namely (a) Terpin Hydrate and Dextromethorphan Elixir and (b) Terpin Hydrate and Codeine Elixir.



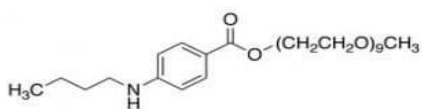
Common side effect includes digestion problem such as nausea, vomiting and constipation.

CENTRALLY ACTING ANTITUSSIVE AGENTS

Centrally acting antitussive agents are medications designed to suppress cough reflex by acting on the central nervous system, primarily in the brain. They are commonly used to alleviate persistent, non-productive coughs that can be caused by various conditions such as respiratory infections or allergies. These agents work by inhibiting the cough center in the medulla oblongata, thereby reducing the urge to cough. Common examples include dextromethorphan and codeine. While effective for short-term use, they are typically not recommended for productive coughs, as this can lead to the accumulation of secretions in the airways.

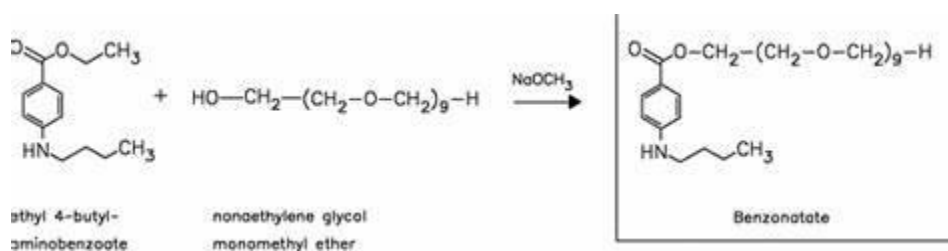
Examples: Benzonatate, Carbetapentane citrate, Noscapine, Levopropoxyphene, Dextromethorphan hydrobromide, Pholcodine

A. *Benzonatate*



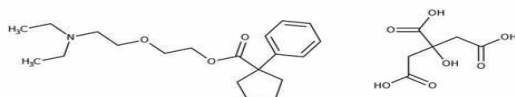
Benzonatate is a non-opioid antitussive agent primarily used to relieve cough symptoms. It works peripherally by anesthetizing the stretch receptors in the lungs and pleura, effectively suppressing the cough reflex. Classified as an ester, its structure is related to procaine, and it is typically administered in capsule form. Benzonatate's pharmacokinetics show rapid absorption,

with peak plasma concentrations reached in about 1-2 hours after oral administration. The drug is metabolized in the liver and excreted via urine. Common side effects may include dizziness, headache, and gastrointestinal disturbances. Due to its mechanism of action, it does not possess the addictive properties associated with opioid cough suppressants, making it a suitable option for managing cough without the risks of dependency.



Its mechanism of action involves the anesthetization of stretch receptors located in the airways and lungs, as well as in the pleura. Specifically, when benzonatate is swallowed, it is absorbed and then distributed to these receptors. By binding to and inhibiting these sensory neurons, it reduces the cough reflex response that is triggered by irritation or inflammation in the respiratory tract. This peripheral action helps suppress the urge to cough without affecting the central nervous system, distinguishing it from traditional opioids.

B. *Carbetapentane citrate*



Carbetapentane citrate is an antitussive agent used primarily to relieve coughs. It works by suppressing the cough reflex in the brain, making it useful in treating cough associated with respiratory conditions. Its chemical structure allows for effective penetration of the central

nervous system, enhancing its efficacy. As a citrate salt, it may also exhibit improved solubility and absorption properties.

1-phenylcyclopentane carbonyl chloride + 2-diethylaminoethoxy ethanol

Condensed 

HCL

Carbetapentane base

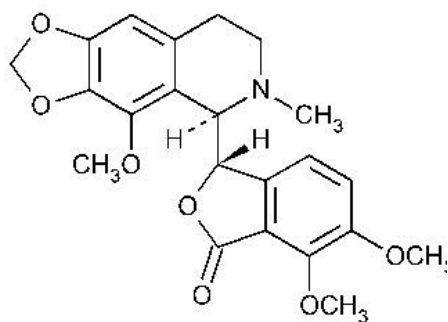
C₂H₅OH 

Citric acid

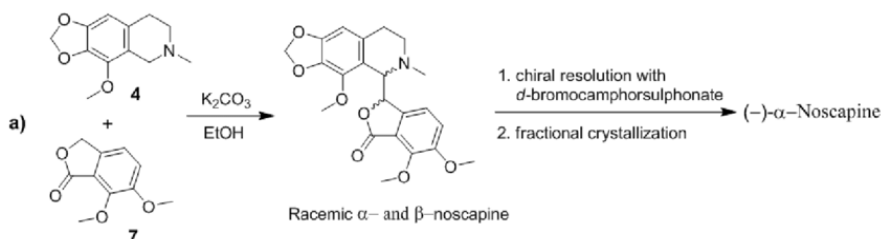
Carbetapentane citrate

Carbetapentane citrate exerts its antitussive effects primarily through its central action on the cough reflex mechanism located in the medulla oblongata of the brain. By interacting with specific receptors in the central nervous system, it inhibits the cough reflex, thereby reducing the urge to cough. Unlike some other cough suppressants that may have a direct effect on the airways, carbetapentane citrate acts centrally, providing relief from dry and persistent coughs that can result from irritants or respiratory conditions. This suppression of the cough reflex helps alleviate discomfort and improves the quality of life for patients experiencing acute or chronic cough, without significantly affecting respiratory function. Furthermore, its formulation as a citrate salt enhances its solubility, potentially improving its absorption and overall efficacy in clinical settings.

C. Noscapine

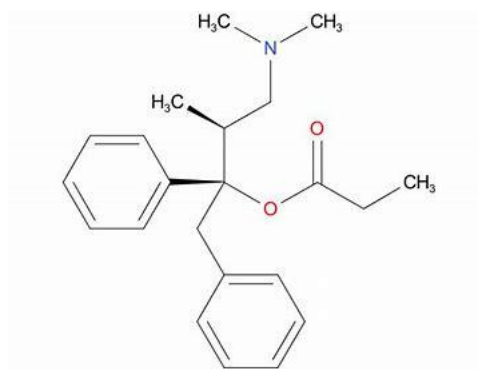


Noscapine is a natural alkaloid extracted from the opium poppy (*Papaver somniferum*) that has garnered attention in medicinal chemistry due to its diverse pharmacological properties. It is primarily known for its antitussive (cough-suppressing) effects, but recent research has highlighted its potential as an anticancer agent. Noscapine acts through various mechanisms, including the inhibition of microtubule assembly, which interferes with cancer cell proliferation and induces apoptosis. Unlike traditional opiates, noscapine does not exhibit significant analgesic properties or addictive potential, making it a candidate for safer therapeutic applications.

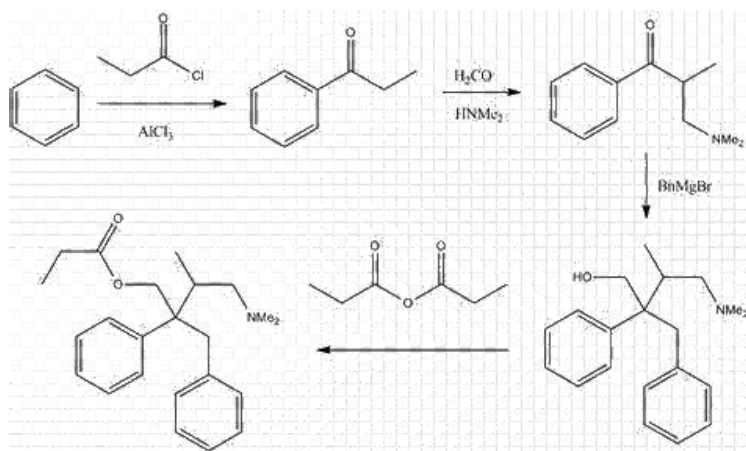


Noscapine acts as an antitussive primarily through its effects on the cough reflex pathway in the central nervous system. It exerts its action by inhibiting the cough center located in the medulla oblongata of the brain, effectively reducing the sensitivity to stimuli that trigger cough. Unlike traditional narcotic cough suppressants, noscapine does not possess significant analgesic or sedative properties, making it an attractive option for patients requiring relief from coughing without the risk of addiction or respiratory depression. Additionally, noscapine may influence other neurotransmitter systems, such as serotonin and norepinephrine, which could further contribute to its ability to alleviate cough by modulating the neural pathways involved in cough reflex.

D. Levopropoxyphene

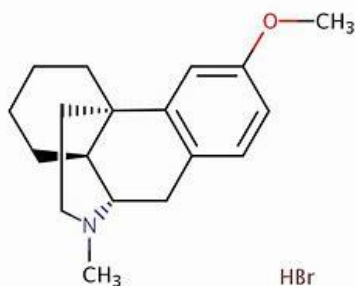


Levopropoxyphene is a synthetic opioid derivative primarily used as an antitussive (cough suppressant). It is structurally related to the opioid analgesic propoxyphene but is specifically designed to reduce cough reflex without significant analgesic properties. Levopropoxyphene acts on the central nervous system by binding to opioid receptors, inhibiting the cough reflex in the medulla oblongata. While effective in treating cough, its use is limited due to potential side effects typical of opioids, including sedation and the risk of dependency. Due to these concerns, it has largely been replaced by safer alternatives in modern medicine.

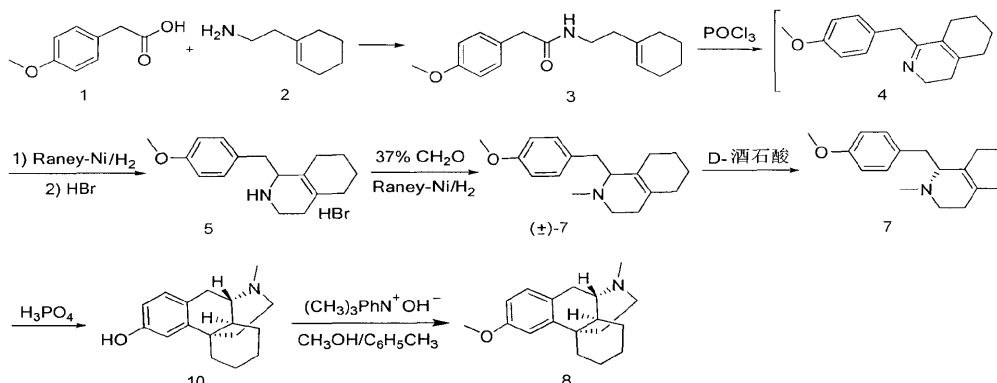


Levopropoxyphene exerts its antitussive effect primarily through its action on the central nervous system, specifically by binding to mu-opioid receptors in the brain, particularly in the cough center located in the medulla oblongata. By activating these receptors, levopropoxyphene modulates the neurotransmission associated with the cough reflex, thereby suppressing the urge to cough. Unlike traditional opioid analgesics that primarily alleviate pain, levopropoxyphene is designed to selectively inhibit the cough response without producing significant analgesic effects. This targeted mechanism allows it to effectively reduce coughing associated with various respiratory conditions. However, its use may be associated with side effects typical of opioid medications, including sedation, which necessitates careful consideration in clinical settings.

E. Dextromethorphan hydrobromide



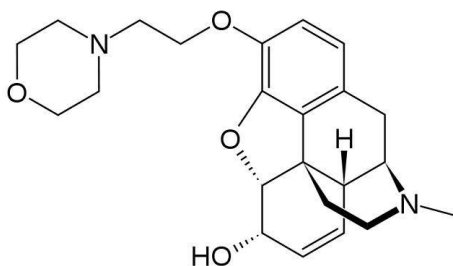
Dextromethorphan hydrobromide is a synthetic derivative of morphine and is primarily used as a cough suppressant in over-the-counter medications. It functions by acting on the central nervous system, specifically targeting the sigma-1 receptor and the NMDA receptor, which helps reduce the urge to cough. Unlike other opioids, it does not produce analgesic effects or significant sedation at standard doses, making it safer for use as a cough suppressant. In medicinal chemistry, it is studied for its metabolic pathways, potential neurological applications, and effects on mood and cognition, along with its safety profile and interactions with other medications.



Dextromethorphan hydrobromide exerts its cough-suppressing effects primarily through its action on the central nervous system, specifically by inhibiting the action of the cough reflex in the medulla oblongata. It is known to interact with the sigma-1 receptor, which is thought to play a role in its antitussive properties. Additionally, dextromethorphan acts as a non-competitive antagonist of the N-methyl-D-aspartate (NMDA) receptor, which modulates glutamate

neurotransmission and may contribute to its effects on mood and cognition. Importantly, while it is related to opioids, dextromethorphan does not produce the same analgesic or sedative effects, as it does not significantly bind to opioid receptors. The drug is metabolized primarily by cytochrome P450 2D6, leading to various metabolite profiles that can influence its efficacy and safety, particularly in individuals with differing metabolic capacities. Overall, its unique mechanism allows for effective cough suppression with a lower risk of addiction and abuse compared to traditional opioids.

F. Pholcodine



Pholcodine is a centrally acting opioid cough suppressant used in medicinal chemistry. It is primarily employed in the formulation of over-the-counter cough medications, where it serves to alleviate dry, non-productive coughs by suppressing the cough reflex in the brain. Unlike traditional opioids, pholcodine has a lower potential for addiction and respiratory depression, making it a safer alternative in certain contexts. Its pharmacological profile involves mu-opioid receptor agonism, but it lacks significant analgesic properties.

The synthesis of pholcodine typically involves a multi-step organic reaction process starting from simple aromatic compounds. One common approach is to use phenol as a starting material, which is first alkylated to form a substituted phenol. This substituted phenol undergoes a series of chemical reactions including methylation and the formation of an ether linkage. The key steps often involve the use of amino and hydroxyl groups, leading to the formation of the morphinan skeleton, which is characteristic of pholcodine. Finally, the product is purified through recrystallization or other separation techniques.

Pholcodine exerts its cough-suppressing effects primarily through its action as a selectively centrally acting opioid. It binds to mu-opioid receptors in the brain, particularly in the medulla oblongata, which is responsible for the cough reflex. By inhibiting neuronal activity in this area, pholcodine reduces the sensitivity of the cough center to peripheral stimuli, effectively suppressing the urge to cough. Unlike other opioids, pholcodine is characterized by a lower affinity for opioid receptors, which contributes to its reduced potential for addiction and respiratory depression. Additionally, it does not exhibit significant analgesic properties, making it a more favorable option for treating coughs without the risks associated with stronger opioids. Overall, pholcodine's unique mechanism allows it to provide symptomatic relief for dry coughs while minimizing adverse effects commonly associated with traditional opioid medications.